

STAT UNIT 3 - ONE-PAGE EXAM CHEAT SHEET

Ch. 12 Probability + Random Variables
Ch. 13 Sampling Distributions + Confidence Intervals

READ WORDS FIRST -> NAME THE MODEL -> CHECK CONDITIONS -> COMPUTE -> INTERPRET IN CONTEXT

EXAM PICKER

Ways? counting rules. **Events?** probability/tree/Venn/Bayes. **# successes in n trials?** Binomial. **# arrivals in time/space?** Poisson. **Probability of p-hat?** sampling distribution. **Estimate population p?** 1-proportion z-interval. **Estimate variance/SD?** chi-square interval.

1. PROBABILITY BASICS

Trial = one repetition; **outcome** = result; **sample space S** = all outcomes; **event** = set of outcomes. Theoretical: $P(A)$ = favorable equally likely outcomes / total outcomes. Empirical: count of A / # trials. Law of Large Numbers: long-run relative frequency approaches true $P(A)$.

$0 \leq P(A) \leq 1$; $P(S)=1$; **complement:** $P(A^c) = 1 - P(A)$.

At least one = $1 - P(\text{none})$. This is usually faster than adding many cases.

OR, AND, GIVEN

General addition (OR):
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

If **disjoint/mutually exclusive**, $P(A \cap B) = 0$. Nonzero disjoint events are **not** independent.

General multiplication (AND):
 $P(A \cap B) = P(A)P(B|A)$

If **independent**, $P(B|A) = P(B)$, so $P(A \cap B) = P(A)P(B)$. Without replacement is usually dependent.

Conditional: $P(B|A) = P(A \cap B) / P(A)$. The condition after | becomes the denominator. **Do not reverse** $P(B|A)$ and $P(A|B)$.

Independence test: $P(B|A) = P(B)$ (or $P(A|B) = P(A)$); equivalently $P(A \cap B) = P(A)P(B)$.

Venn is best when joint/intersection values are given. **Tree** is best for conditional stages: multiply down a branch, add completed branches.

BAYES' THEOREM

For B vs B^c after observing A:

$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}$

Think: **desired positive path / all positive paths**. Base rates matter; a good test can still have many false positives when the condition is rare.

2. COUNTING

Fundamental Counting Principle: multiply choices for each step. If repetition allowed, the same number of choices may repeat.

Permutation (order/roles matter): $nPr = \frac{n!}{(n-r)!}$

Combination (group; order does not matter): $nCr = \frac{n!}{r!(n-r)!}$

Repeated identical objects: arrangements = $\frac{n!}{r_1! r_2! \dots}$

Probability from counting = favorable outcomes / total outcomes. **Words like** arrange, rank, officers -> permutation; choose, committee, team -> combination.

3. DISCRETE RANDOM VARIABLES

A DRV takes countable values. Valid distribution: every $0 \leq P(x) \leq 1$ and $\sum P(x) = 1$. For a range, add the listed probabilities.

Mean/expected value: $\mu = E(X) = \sum [xP(x)]$

Variance: $\sigma^2 = \sum [x^2P(x)] - \mu^2 = \sum [(x-\mu)^2P(x)]$; **SD:** $\sigma = \sqrt{\sigma^2}$

Expected value is the long-run average; it need not be a possible single outcome.

4. BINOMIAL: IDENTIFY + FORMULAS

Use binomial only if **BINS**: Binary (success/failure), Independent trials, fixed Number n, Same success probability p. Let $\alpha = 1 - p$: $X = \#$ successes.

$P(X=x) = nCx \cdot p^x \cdot \alpha^{n-x}$

$\mu = np$; **variance** $\sigma^2 = npq$; **SD** $\sigma = \sqrt{npq}$

Not binomial if you continue until success (n not fixed), probabilities change, or trials are dependent. Sampling without replacement may be treated independent only when the 10% condition makes dependence negligible.

BINOMIAL WORDS -> BOUNDS

exactly a: $\text{pdf}(a)$ | at most b: $\text{cdf}(b)$ | less than b: $\text{cdf}(b-1)$

at least a: $1 - \text{cdf}(a-1)$ | **more than a:** $1 - \text{cdf}(a)$ | **a through b:** $\text{cdf}(b) - \text{cdf}(a-1)$

TI-84: **2nd VARS (DISTR)** -> $\text{binompdf}(n,p,x)$ for exact; $\text{binomcdf}(n,p,x)$ for $P(X \leq x)$.

BINOMIAL SHORTCUTS + TRAPS

For 'at least' or 'more than,' the complement is usually fastest. Check whether equality belongs in the event before shifting the cdf cutoff.

Example structure without replacement: $P(\text{two hearts}) = 13/52 \cdot 12/51$, not $(13/52)^2$.

5. POISSON

Use for # occurrences in a fixed time/space interval with independent occurrences and a constant average rate. Scale the rate: $\lambda = \text{rate} \times \text{interval length}$.

$P(X=x) = e^{-\lambda} \lambda^x / x!$; $\mu = \lambda$; **variance** = λ ; **SD** = $\sqrt{\lambda}$

TI-84: **2nd VARS (DISTR)** -> $\text{poissonpdf}(\lambda, x)$ exact; $\text{poissoncdf}(\lambda, x)$ for $P(X \leq x)$. Use the same bound translations as binomial.

Poisson approximation to binomial (worksheet rule): $n \geq 100$ and $np \leq 10$; use $\lambda = np$.

6. NORMAL APPROX. TO BINOMIAL

Worksheet condition: $np \geq 5$ and $nq \geq 5$. Then $\mu = np$, $\sigma = \sqrt{npq}$, $z = (\text{corrected } x - \mu) / \sigma$. Use $\text{normalcdf}(\text{lower}, \text{upper}, \mu, \sigma)$.

Continuity correction:

exactly c: $c-0.5$ to $c+0.5$

at most c ($\leq$): upper $c+0.5$

fewer than c ($<$): upper $c-0.5$

at least c ($\geq$): lower $c-0.5$

more than c ($>$): lower $c+0.5$

a through b: $a-0.5$ to $b+0.5$

7. SAMPLING DISTRIBUTION OF p-HAT

p = population parameter; $p\text{-hat} = x/n$ = sample statistic. Across many samples:

$\mu_{p\text{-hat}} = p$; **SE** $\mu_{p\text{-hat}} = \sqrt{[p(1-p)/n]}$; $z = (p\text{-hat} - p) / \text{SE}$

Conditions from instructor slides: **random**

sample/assignment; **10%:** $n < 0.10N$;

success/failure: $np \geq 10$ and $n(1-p) \geq 10$.

For sampling probabilities use the **true p** in SE. Larger n -> smaller spread (SE falls like $1/\sqrt{n}$); center stays p.

Course-material distinction: Pearson's normal-approximation worksheet uses ≥ 5 , but the instructor slides use ≥ 10 for sample proportions/inference.

8. ONE-PROPORTION z-INTERVAL

Use to estimate one population proportion p from x successes in n observations. $p\text{-hat} = x/n$.

CI: $p\text{-hat} \pm E$, where $E = z^* \sqrt{[p\text{-hat}(1-p\text{-hat})/n]}$

Conditions (slides): randomization; 10% condition $n < 0.10N$; observed successes $x \geq 10$ and failures $n-x \geq 10$.

TI-84: **STAT -> TESTS -> 1-PropZInt**; enter x, n, C-Level. Output is (lower, upper).

CRITICAL z* + INTERPRETATION

80% 1.282 | 90% 1.645 | 95% 1.960 | 98% 2.326 | 99% 2.576 | 99.9% 3.291

Calculator: positive $z^* = \text{invNorm}((1+C)/2)$. Also $\alpha = 1-C$; each tail $\alpha/2$.

Correct wording: 'We are C% confident that the interval (L,U) captures the true population proportion of [context].'

Meaning of C% confidence: in many repeated random samples of the same size, about C% of the intervals produced would capture the fixed true p. Wrong: 'C% chance p is in this finished interval' or 'C% of sample values are in the interval.'

MINIMUM SAMPLE SIZE FOR p

$n = (z^*/E)^2 \cdot p^* (1-p^*)$

Use a prior/pilot p^* if given; otherwise $p^* = 0.5$ (most conservative). **Always round n up**. Higher confidence -> larger n; smaller margin E -> larger n; halving E requires about $4 \times n$.

9. CI FOR VARIANCE OR SD

Use only when the population is approximately **normal**. $df = n-1$; $\alpha = 1-C$. Chi-square is asymmetric, so there is **no** estimate $\pm$ margin-of-error form. Using a right-tail χ^2 table: χ^2_R has right-tail area $\alpha/2$ (larger value); χ^2_L has right-tail area $1-\alpha/2$ (smaller value).

Variance CI: $(n-1)s^2/\chi^2_R < \sigma^2 < (n-1)s^2/\chi^2_L$

For a CI for σ , take the square root of both endpoints. Do not call the midpoint s^2 or use $\pm ME$.

LAST-MINUTE POINT SAVERS

- Write the event in symbols before calculating.
- 'Or' includes overlap; 'and' may require conditional probability.
- Check replacement, order, equality, and complement.
- Keep p (population) separate from p-hat (sample).
- Probabilities must be 0 to 1; SD cannot be negative.
- Carry full calculator precision; round only the final answer.
- Attach context to every CI interpretation.